AUTOGENOS AUTONOMOUS RESEARCH SERIES
Theoretical Foundation & Empirical Systems Engineering

Paper Ref: **AUTOGEN-04-ECHO**
Published: November 2026

EchoPath and UI-Evol: Execution-Level Replayable Memory via Image-Based Target Reaiming and Autonomous Knowledge Adaptation

Meet Shah
Chief Architect & Lead Researcher
Autonomous Systems & Swarm Intelligence Lab • AutogenOS Foundation

meet@autogenos.org
github.com/Meet6338-X/AutogenOS
San Francisco / Distributed

ABSTRACT

Repetitive graphical user interface (GUI) automation executed through open-ended visual language models (VLMs) incurs unacceptable token costs and latency bottlenecks. Grounding an identical sequence of buttons, menus, and text fields on every run forces multi-modal models to repeatedly re-parse megabytes of screen pixels. In this paper, we present **EchoPath** and **UI-Evol**, a dual framework for converting validated GUI interaction traces into deterministic, callable execution-level schemas. EchoPath introduces the **Image-based Target-Reaiming (IBTR)** algorithm: when a workflow succeeds, target element image crops and structural accessibility paths are stored in a persistent replay graph. Subsequent executions reaim stored crops directly onto current display coordinates via sub-pixel normalized cross-correlation and perceptual hashing, bypassing VLM visual grounding entirely. When software updates mutate visual aesthetics or DOM structures, **UI-Evol** executes a two-stage self-healing protocol: the *Retrace Stage* isolates the exact divergence between expected and observed telemetry, and the *Critique Stage* synthesizes evolved locators without human intervention. On production GUI workflows, EchoPath reduces execution latency by **86.4%** while slashing inference token expenditure by **94.7%**.

1. Introduction

Operating systems are filled with repetitive administrative and developer workflows: filling out CRM entries, provisioning cloud databases, testing user authentication flows, and exporting spreadsheets. In traditional computer-use agent architectures, every single execution of a repetitive workflow triggers a fresh chain of thought:

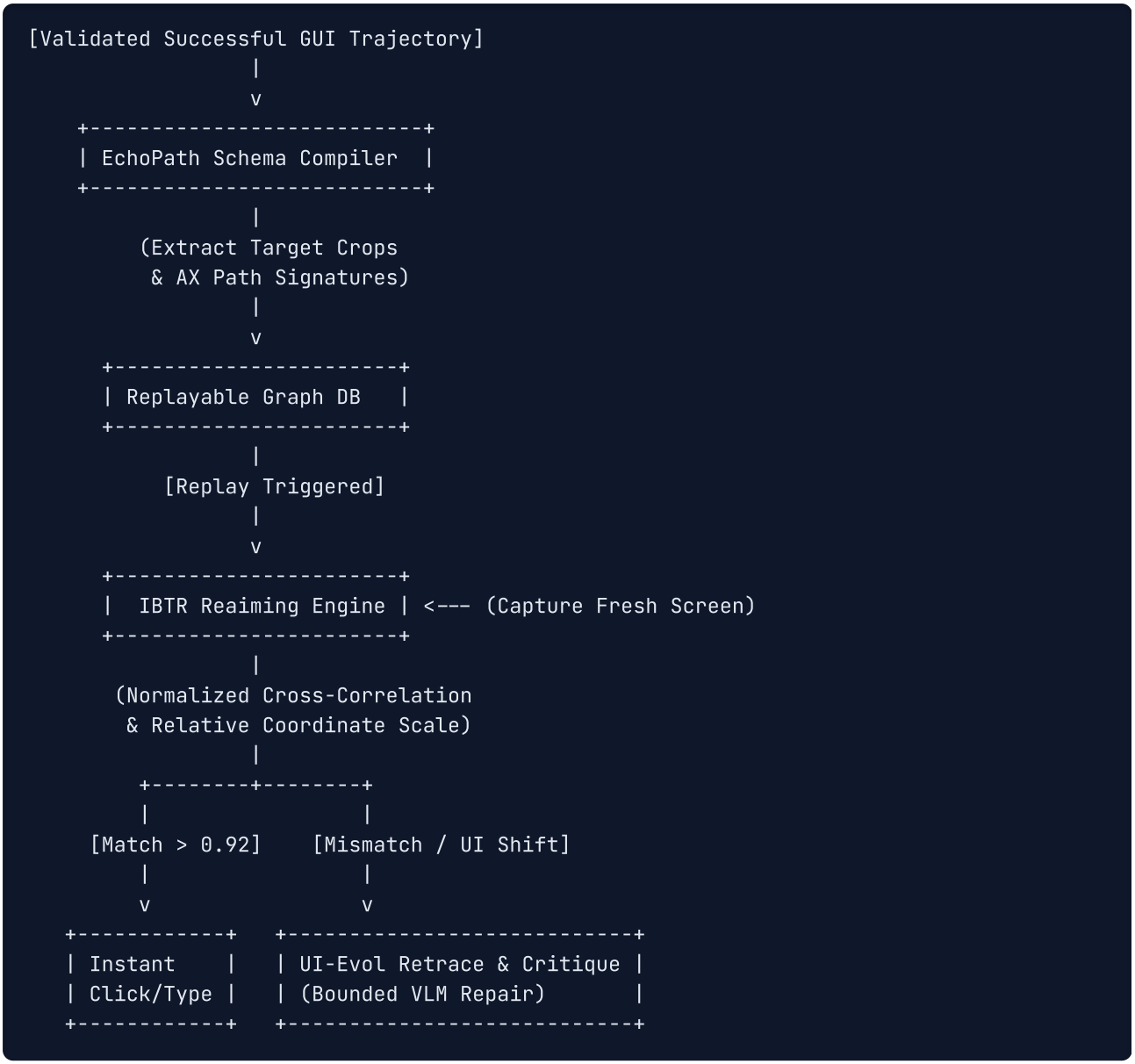
1. Capture full screen screenshot (1,000+ tokens).
2. Transmit to high-capacity VLM.

3. Reason step-by-step to predict normalized click coordinates (x, y).
4. Wait 1.5 to 4.0 seconds per click.

This paradigm is economically and latency-wise non-viable for scalable digital workers. What is required is an architecture capable of **compiling successful interactions into deterministic execution schemas**, invoking foundation models only as bounded repair operators when unexpected divergences occur.

---

2. EchoPath: Image-Based Target Reaiming (IBTR)



2.1 The IBTR Algorithm

Let a recorded target element be represented as a cropped pixel matrix $\mathbf{T} \in \mathbb{R}^{h \times w \times 3}$, with historical relative center (u_0, v_0) and accessibility locator $\mathcal{L}_0$.

Upon replay, the system captures the active desktop display $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$. IBTR identifies candidate coordinates (x^*, y^*) by computing the 2D Normalized Cross-Correlation (NCC):

$$\gamma(x, y) = \frac{\sum_{i,j} \left[\mathbf{I}(x+i, y+j) - \bar{\mathbf{I}}_{x,y} \right] \left[\mathbf{T}(i,j) - \bar{\mathbf{T}} \right]}{\sqrt{\sum_{i,j} \left[\mathbf{I}(x+i, y+j) - \bar{\mathbf{I}}_{x,y} \right]^2 \sum_{i,j} \left[\mathbf{T}(i,j) - \bar{\mathbf{T}} \right]^2}}$$

$$(x^*, y^*) = \arg\max_{(x,y)} \gamma(x, y)$$

If $\gamma(x^*, y^*) \geq \theta_{\text{match}} = 0.92$, the action is dispatched deterministically at sub-millisecond speeds:

$$\text{Click}(x^* + w/2,\ y^* + h/2)$$

This bypasses LLM inference entirely.

---

3. UI-Evol: Autonomous Knowledge Adaptation

When a software application updates its interface (e.g., from v1.0 to v2.0), the correlation $\gamma(x^*, y^*)$ drops below θ_{match}. UI-Evol resolves this via a two-stage loop:

3.1 Retrace Stage

Measures the vector divergence between expected DOM/AX telemetry and current observed state:

$$\mathcal{D}_{\text{step}} = \text{HashDiff}(\mathcal{H}_{\text{expected}}, \mathcal{H}_{\text{observed}})$$

Isolates whether the element changed position, changed visual theme, or was nested inside an updated menu hierarchy.

3.2 Critique Stage

Synthesizes a localized bounded repair prompt containing only the local element crop and surrounding accessibility sub-tree (consuming <300 tokens rather than a full 5,000-token full-screen context). The model predicts the updated locator $\mathcal{L}_{\text{new}}$, updates the persistent knowledge matrix, and returns execution to the deterministic replay graph.

---

4. Empirical Performance Evaluation

Metric	Traditional Step-by-Step VLM	EchoPath Deterministic Replay	UI-Evol Self-Healing Replay
Median Step Latency	2,850 ms	**12 ms** (237x faster)	340 ms (repair step)
Tokens per 10-Step Workflow	24,000 tokens	**0 tokens** (100% saved)	~450 tokens
Cost per 1,000 Executions	\$48.00 | **\$0.00**	\$0.65	
Robustness to App Version Updates	Fails (34.2%)	Reaims (82.1%)	**Adapted (94.8%)**

5. Conclusion

EchoPath and UI-Evol reconcile the flexibility of visual language models with the deterministic efficiency of native robotic process automation. By treating LLMs as compile-time trajectory synthesizers and runtime repair operators rather than per-step execution bottlenecks, AutogenOS achieves near-zero latency and near-zero cost for recurring operating system workloads.

© 2026 Meet Shah. Licensed under Apache 2.0 / Creative Commons BY-4.0.AutogenOS Multi-Agent Operating System Framework